

RAFHA, Saudi Arabia

WMO: 403620

Lat: 29.626N

Lon: 43.491E

Elev: 449

StdP: 96.04

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.6	3.2	-11.9	1.4	11.2	-9.0	1.9	13.1	11.9	13.4	10.5	13.5	1.6	300	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.6	45.2	20.7	44.2	20.6	43.1	20.4	23.9	40.9	22.7	40.5	21.9	39.6	4.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.7	32.0	16.8	12.7	30.4	15.2	11.4	28.8	74.5	41.0	69.6	40.4	66.1	39.3	33.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.6	8.5	DB	-1.5	46.9	2.7	1.0	-3.4	47.6	-5.0	48.3	-6.5	48.8	-8.5	49.6	(4)
(5)			WB	-3.6	25.0	2.3	3.3	-5.2	27.4	-6.6	29.4	-7.9	31.2	-9.6	33.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	24.1	11.4	13.6	18.3	23.7	29.6	33.1	34.7	35.3	32.1	26.8	17.8	12.8	(6)
(7)		DBStd	9.21	3.34	3.66	3.95	3.76	3.00	2.17	2.10	2.17	2.55	3.39	3.88	3.62	(7)
(8)		HDD10.0	46	22	9	1	0	0	0	0	0	0	0	1	13	(8)
(9)		HDD18.3	640	217	138	50	3	0	0	0	0	0	1	56	176	(9)
(10)		CDD10.0	5211	64	109	257	412	608	692	767	784	663	520	234	99	(10)
(11)		CDD18.3	2762	1	4	48	166	349	442	509	526	413	262	39	3	(11)
(12)		CDH23.3	43778	14	63	572	2008	5229	7495	8909	9269	6590	3239	352	37	(12)
(13)	CDH26.7	29475	1	13	219	976	3292	5252	6483	6840	4467	1814	113	4	(13)	
(14)	Wind	WSAvg	3.8	3.7	3.9	4.3	4.4	4.0	3.9	4.0	3.4	3.3	3.7	3.5	3.4	(14)
(15)	Precipitation	PrecAvg	77	14	8	12	11	4	0	0	0	4	16	8	(15)	
(16)		PrecMax	133	50	32	97	64	27	3	0	1	0	20	70	46	(16)
(17)		PrecMin	8	0	0	0	0	0	0	0	0	0	0	0	0	(17)
(18)		PrecStd	39	14	10	19	14	6	1	0	0	0	6	18	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	28.8	35.5	39.0	42.9	45.2	46.8	44.8	40.8	32.6	27.1	(19)	
(20)			MCWB	13.9	15.0	17.4	19.0	19.6	19.7	21.8	20.9	20.6	18.8	16.8	14.1	(20)
(21)		2%	DB	22.0	25.0	31.8	36.1	40.9	43.3	45.2	45.5	43.1	38.1	29.9	23.8	(21)
(22)			MCWB	12.7	13.5	15.8	18.0	19.7	20.2	20.7	20.8	20.4	18.7	16.0	13.1	(22)
(23)		5%	DB	19.8	22.8	28.9	34.1	39.1	42.2	44.0	44.5	41.7	36.8	27.2	21.2	(23)
(24)			MCWB	11.5	12.1	14.8	17.6	19.2	19.9	20.4	20.9	19.8	18.5	15.0	12.5	(24)
(25)	10%	DB	17.5	20.6	26.2	32.1	37.9	41.0	42.8	43.2	40.1	34.9	25.0	19.1	(25)	
(26)		MCWB	10.5	11.6	13.9	16.7	18.9	19.8	20.1	20.8	19.4	17.8	14.4	11.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.7	17.6	19.0	22.9	23.8	24.7	24.5	25.4	24.5	21.9	19.5	16.5	(27)
(28)			MCD B	21.7	24.8	31.7	32.0	37.8	40.0	42.6	42.9	41.0	35.4	26.0	22.6	(28)
(29)		2%	WB	14.0	15.2	17.2	20.2	22.0	22.6	23.1	23.7	22.6	20.4	17.8	15.0	(29)
(30)			MCD B	19.5	21.4	27.8	31.4	36.7	40.3	42.2	41.8	39.3	34.2	24.2	20.3	(30)
(31)		5%	WB	12.7	13.7	15.8	18.9	20.6	21.4	22.1	22.6	21.1	19.4	16.8	13.7	(31)
(32)			MCD B	17.7	19.8	26.4	30.7	35.6	39.6	41.1	41.6	38.5	33.3	23.2	19.3	(32)
(33)	10%	WB	11.4	12.3	14.5	17.7	19.5	20.4	21.0	21.5	19.9	18.4	15.7	12.4	(33)	
(34)		MCD B	16.5	19.0	24.8	29.2	35.3	38.9	40.2	41.1	37.9	31.9	22.7	18.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.0	12.0	13.2	13.6	14.1	15.3	15.4	15.6	15.2	13.9	11.6	11.0	(35)
(36)			MCDBR	13.8	14.9	16.1	15.8	15.9	16.3	16.1	16.0	16.1	15.9	14.8	13.6	(36)
(37)			MCWBR	7.6	7.7	7.7	6.9	6.7	6.8	6.5	6.4	6.8	7.1	6.7	7.0	(37)
(38)		5% WB	MCD BR	12.0	12.5	14.6	14.2	14.6	15.6	15.8	15.9	15.9	14.5	11.5	11.2	(38)
(39)			MCWBR	7.2	7.5	7.8	7.2	7.3	7.8	7.6	7.8	8.0	7.2	6.0	6.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.411	0.451	0.515	0.610	0.632	0.560	0.531	0.543	0.514	0.540	0.477	0.430	(40)	
(41)		taud	1.956	1.852	1.708	1.555	1.531	1.654	1.735	1.715	1.759	1.709	1.824	1.920	(41)	
(42)		Ebn at Noon	792	796	773	718	705	754	774	760	764	700	712	744	(42)	
(43)		Edh at Noon	154	187	231	279	287	252	232	234	217	214	175	153	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.77	4.71	5.78	6.56	7.23	8.04	7.76	7.23	6.30	5.00	3.86	3.44	(44)	
(45)		RadStd	0.22	0.22	0.22	0.23	0.28	0.19	0.24	0.20	0.21	0.20	0.32	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.73	N/A	-2.27	+1.06	-0.93	N/A	N/A	-72	+336	+256	(46)	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)	

Nomenclature: See separate page